

B.Tech. Degree II Semester Regular and Supplementary/ I Semester Supplementary Examination May 2017

CS/EC/IT GE 15-1203 B & CE/EE/ME/SE GE 15-1103 A
ENGINEERING MECHANICS
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

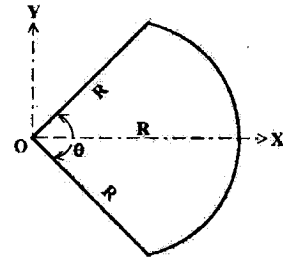
PART A

(Answer *ALL* questions)

(Any missing data, if any, may be suitably assumed)

(10 × 2 = 20)

- I. (a) What is a force? Explain types of forces with examples.
 (b) Show that angle of friction and angle of repose is same for given surfaces of contact.
 (c) State and explain Pappus theorems.
 (d) Determine the moment of inertia of a sector of a circle of radius R about Y-axis.

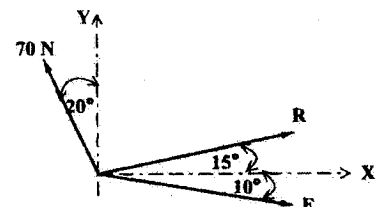


- (e) What is product of inertia? When will product of inertia of a given section be a minimum?
 (f) Differentiate between method of joints and method of sections in the analysis of plane trusses.
 (g) The acceleration of a particle is given by $a = t^3 - 2t^2 + 4$, where 't' is in seconds. The particle after travelling for 2 seconds attains a velocity of 8 m/s, and covers a distance of 15 m. Determine the equations of velocity and displacement.
 (h) A sphere of mass M dropped vertically from a height 'h' on to the horizontal floor and raised to a height of 0.8h after one bounce. Determine coefficient of restitution.
 (i) What is a compound pendulum? Explain.
 (j) State and explain the principle of angular momentum.

PART B

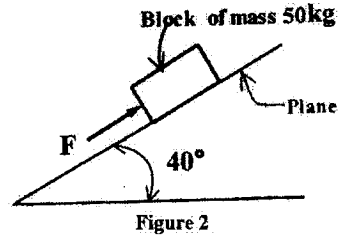
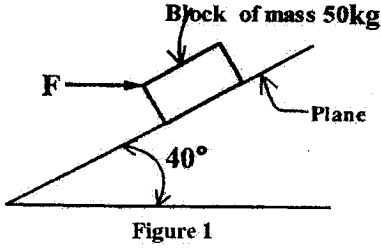
(4 × 10 = 40)

- II. (a) Determine the magnitude of force F for the resultant R as shown in figure to be at an angle of 15° with reference to X-axis. Also determine the resultant R.



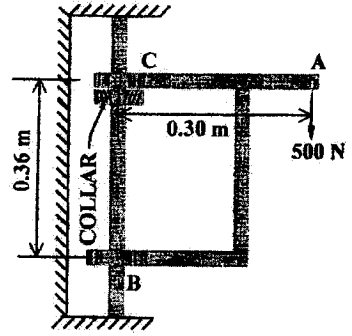
(P.T.O.)

- (b) Determine the magnitude of force F to be applied to keep the body in equilibrium and the normal reaction exerted on the plane for the system shown in Figure 1 and Figure 2. Assume surfaces of contact to be smooth. (5)

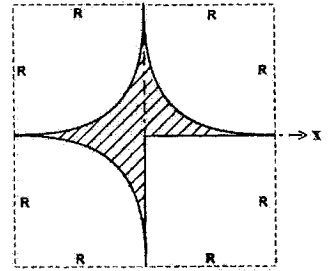


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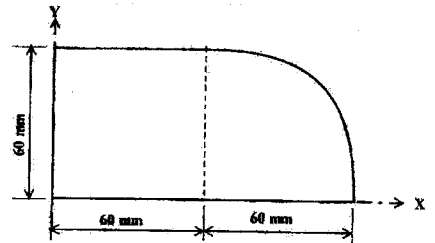
- III. (a) A bracket ACB can slide freely on the vertical shaft BC but is held by a small collar attached to the shaft as shown in the figure. Neglecting friction, determine reactions at B and C for the vertical load. (5)



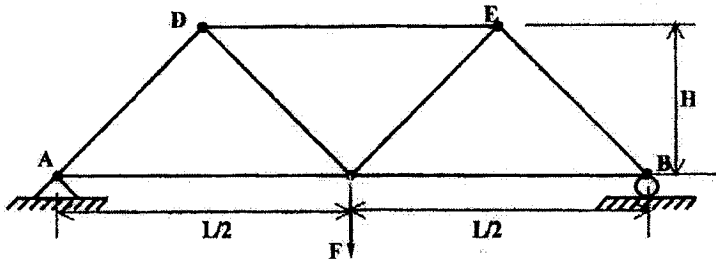
- (b) Determine the centroid of the shaded plane area as shown in figure. (5)



- IV. (a) Determine the product of inertia of the plane section shown in figure. (5)

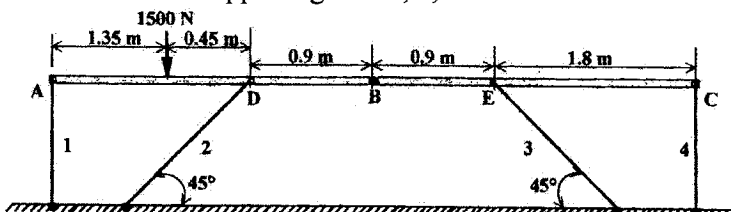


- (b) Applying the principle of virtual work determine the axial force in the member DE of the loaded simple truss shown in figure. (5)

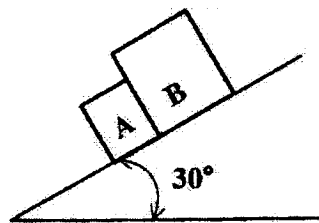


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- V. A compound beam is shown in figure. Determine the force on the connecting pin B, and axial forces in four supporting bars 1, 2, 3 and 4. (10)



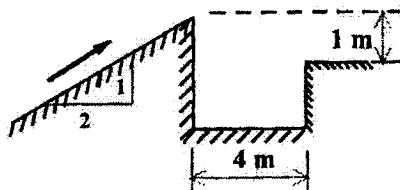
- VI. (a) A stone after falling 4 seconds from rest breaks a glass pane and in breaking it loses 30% of the velocity. How far does it fall in the next second? Take $g = 9.81 \text{ m/s}^2$.
- (b) Blocks A and B are of masses 10 kg and 30 kg respectively are placed in a 30° inclined plane as shown in figure. The coefficient of friction between block A and the plane is 0.25, and between the block B and the plane is 0.15. What is the force between them as they slide down? What is the acceleration of blocks?



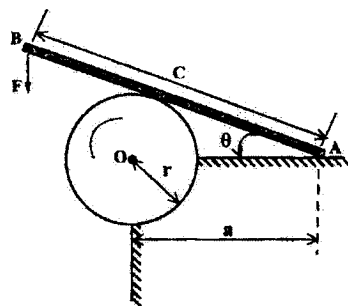
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- VII. (a) A stone is dropped into a well with no initial velocity and 4.5 s later; the sound of impact is heard. Then a second stone is thrown into the well with an initial velocity of V_0 , and the splash is heard 4 s later. If the velocity of sound is 336 m/s, determine the initial velocity of the second stone.
- (b) A bullet moving with a velocity of 330 m/s penetrates through a plank. The bullet has a velocity of 286 m/s, the moment it comes out of the plank. Determine the number of planks through which it penetrates before it comes to rest.

- VIII. (a) A stunt man is to drive an automobile across the ditch shown in figure. Determine the minimum take off velocity of the automobile and its time of flight.

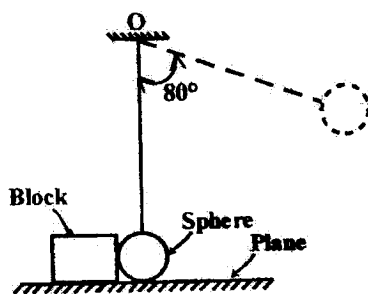


- (b) A right circular drum of radius r and weight W rotating at ω_0 rpm is braked by the device shown in figure. Develop a formula for the time t required to bring the drum to rest if the coefficient of friction between drum and braking bar is μ .

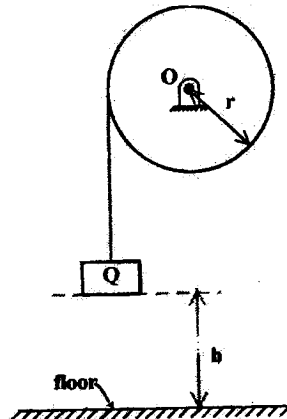


OR

- IX. (a) A sphere of mass 5 kg hung by means of a string 1 m long as shown in figure, is raised by 80° to the right side, and then released from its position. The 5 kg sphere in its downswing hits a block of mass 2.5 kg. Determine the final velocities and loss of kinetic energy. Neglect friction between the block and the plane.



- (b) A solid right circular drum of radius r and weight W is free to rotate about its geometric axis as shown in figure. A flexible cord is wound round the circumference of drum and carries a weight Q at its end. If the weight Q is released from rest :



- (i) Find the time t required to fall it through the height h and
- (ii) With what velocity V , will it strike the floor? Take $r = 0.3 \text{ m}$, $W = 150 \text{ N}$, $Q = 50 \text{ N}$ and $h = 3 \text{ m}$.